

KINDLED

Final Game Design Document

Aaron Low · 27 July 2026 · Unreal Engine 5.8

Supersedes the Assignment #1 first draft (21 July 2026), submitted under the working title *Emberkeep*.

Working title. Named for what happens to the people who gather to you; see §2.3.

You are the only light. Everything that lives, lives inside your circle — and they think you are a god.

1 Executive Summary

Kindled is a single-player roguelike in which you carry the only fire in a pitch-dark world, and an army gathers in your light because outside it they die.

The world is not at night — it is *dark*, and the dark takes what it touches. Power exists only where it passes through fire, so a **flame-bearer** is the one place anything can happen. People gather to you, fight for you, and need you to live; this is not loyalty, it is physics. They do not think you are a person — a bearer is a god to the people around them, and you can neither ask for that nor decline it. Other bearers are scattered in the dark, each with a small congregation in a small circle of light. Alone, a flame holds a room. **United, flames do what no single fire can** — and finding them is what a run is *for*. They are people you locate in the dark and unite with, not other players in a lobby (§2.3).

In play: a top-down roguelike run where you command a growing retinue — three to ten units early, hundreds by late run — through four broad **stances** rather than unit selection or targeting. It renders in a locked four-value 2-bit palette chosen so a several-hundred-unit battle stays legible by silhouette and value alone. Retinue growth, not loot, is the progression axis and the run's reward.

WIN CONDITION — PER RUN

Survive three escalating waves and kill what the last one brings. The fire is still burning when the floor is clear.

LOSS CONDITION

Hero death. The flame goes out, everyone standing in it is in the dark, and the run ends immediately and finally.

The rule the whole design hangs on is the **leash**: every unit has a maximum distance from the bearer, and that distance is drawn on the ground as the reach of your light. Inside it, lit floor and full-value units. Outside, the dark ground state and units dimmed a palette step. A unit that strays too far drops whatever order you gave and returns — not disobedience, but *running back to the light*. You may anchor a line at a chokepoint, but you can never walk away from it.

PROJECT STATUS — PLAYABLE, MEASURED, NOT YET PASSING ITS OWN GATE

The Gate 1 prototype (all four stances, the leash, a three-wave run, win and lose states) was built 2026-07-22 and has been tuned against measured runs since. It is calibrated so that a **zero-input run loses** — narrowly and repeatably, three runs out of three — so everything the player adds is the margin between defeat and victory. The concept's go/no-go gate, entity count at frame budget, is *not* passed: §5 states exactly why and what it costs.

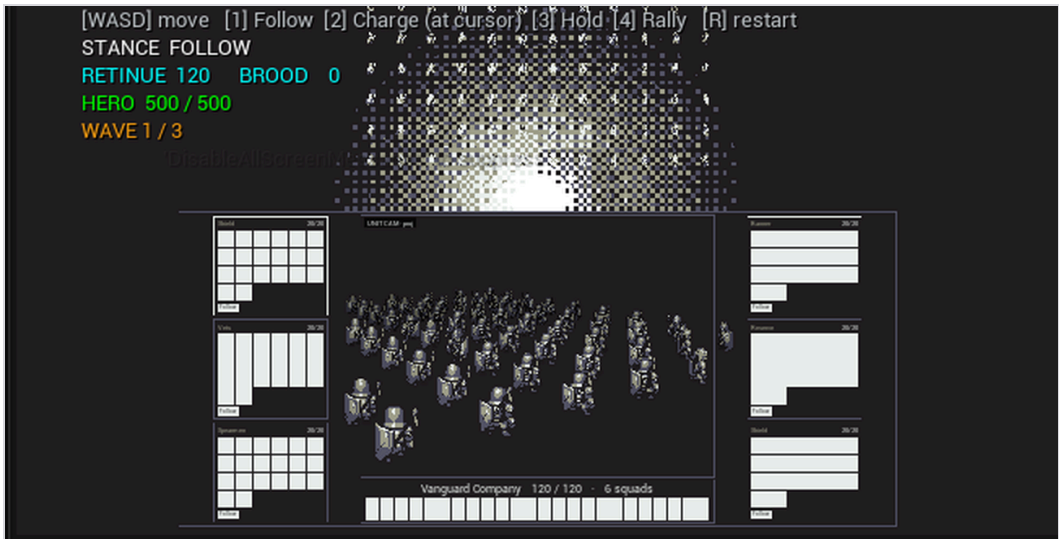


Figure 1 — the game as it runs today. The bearer's flame is the bright core; the dithered falloff around it is the lit floor, which is also the leash radius (§3.2). 120 retinue hold formation inside it while brood approach from the dark at the top of frame. The panel along the bottom is the Unit Cam: a close-up of the line, flanked by six squad-strength readouts. *Debug-box renderer — the sprite path is repaired but unmeasured (§5.1).*

2 What Changed Since the First Draft

Three inputs drove this revision: a narrative reset, three corrections that review and measurement forced, and my own judgement calls. The largest change is first.

2.1 The narrative reset — the whole world was discarded

Assignment #1 described the *Undervault*: a kingdom a thousand years dark, the Hollow Crown, three factions, five named NPCs, fifteen persistent world flags and eight decision-event templates. On 2026-07-22, **all of it was discarded**. The world document is superseded in full and kept unedited as history.

The reason: **the set dressing had outrun the prototype**. Eight event templates written and none built; fifteen world flags specified and one stubbed write planned — while the playable build carried a question no amount of faction lore could answer, namely whether the leash reads as a rule the player understands or whether the army just feels disobedient. The fiction was accumulating faster than the thing it described.

The replacement premise (§1) is deliberately short and mechanics-first, and ships with a rule every future piece of lore must pass: **if a piece of fiction does not explain a mechanic that already exists or is already needed, it is decoration and it waits**. The premise was written against that test:

FICTION	MECHANIC IT EXPLAINS	STATUS IN CODE
Outside the light, the dark takes you	The leash — radius, break latch, warn bit	Built
A god is obeyed in intent, not in orders	Stances; no unit selection, no micro	Built — 4 stances
Fire must be fed or it dims	Upkeep — units degrade rather than die	Specced
You are the only light source	Hero relevance without hero damage	Unblocks the 55-DPS problem
Many bearers, united	The run objective — find the other fires, bring them together	Unbuilt; the run's spine (§2.3)
They die for you gladly	Sacrifice events priced by what it costs <i>you</i>	Specced

What the reset bought immediately

- **The leash became visible instead of a rule.** Rendering the leash radius as the lit floor answers the prototype's open question directly, and costs one floor decal plus a per-unit value shift the renderer already has the data for.
- **Hold and Charge acquired a price.** Both push units toward the edge of the leash — which is precisely where the fiction says it is dangerous to stand. That is not a bug, it is the game. Three of the four stances now have a cost the player can feel; previously they were movement modes.
- **The hero stopped needing damage.** The first draft worried openly that a 55-DPS hero was "a camera with a health bar." The hero is now *the thing everyone is standing in*: abilities concern where the light reaches, how bright, how long, and at what cost.

What the reset cost — stated, not hidden

The first draft's **meta-win condition went with the lore**. Progression across runs was "your decisions write world flags that weaken the Hollow Crown's grip." There is no Hollow Crown and there are no flags. Meta-progression is now *deliberately unwritten*, because the prototype question that decides what a persistent world would need to contain — does standing in the circle stagnate into a dominant strategy? — is unanswered. Writing it now would repeat the mistake the reset exists to undo. Per-run win and loss (\$1) are unaffected and are what the build implements today.

2.2 Three corrections review and measurement forced

The first two contradict sentences in the submitted first draft.

a. The Shield Wall claim was wrong. Assignment #1 said "the Vanguard's Hold is a literal Shield Wall that blocks enemy pathing." A blocking wall needs per-unit pathing obstruction — exactly the individual special-casing Mass Entity rules out at horde scale. **Corrected 2026-07-26 toward the simulation:** Hold pins the formation where it was issued, but nothing about it changes how far an enemy will divert to engage. The tide bites what is near its path and flows past what is not. Shield Wall is now a positioning tool that reads as a wall only where geometry already makes it one.

b. A balance claim I made was false, and measurement caught it. Moving from continuous damage to discrete swings, I wrote that average DPS was unchanged. Wave-1 survivors fell from 97–103 to 60–62. The cause: `MaxAttackersPerUnit` — called "the single most important dial in the model" in my own design — **never bounded a damage rate at all**. It capped the first N enemies in *grid iteration order*, and the grid rebuilds every frame as units move, so that set churned and far more than N attackers landed blows per interval. Fixed with geometry rather than counters: each attacker publishes the distance to its K-th nearest enemy, and a victim takes a blow if inside it — one compare, no cross-entity writes. That fix then exposed a second thing: a *constant* K removed a stabiliser the broken model had by accident, and fights went **bimodal**. Hence the shipped defaults, which restore the stabiliser deliberately.

c. The renderer cannot hold the bar it was assumed to hold. At 1,000 enemies the debug-box renderer costs **14.62 ms on draw alone** against a 16.6 ms frame — and 350 ms at 10,000. Draw time is frame time at every count tested. Since the Spike 1 floor is "1,000+ units at 60 fps," this renderer cannot pass it. Now the project's binding technical fact rather than an assumption (\$5).

2.3 What I changed by my own judgement

- **The title.** *Emberkeep* came from the discarded canon — it named a place, and the place no longer exists. **Kindled** names a *state* instead, and specifically the state of everyone who is not the player: they have been kindled, by you, and they will go out without you. The participle was chosen over the bare verb deliberately. *Kindle* would have named what the bearer does; *Kindled* names what it does to the people standing in the light, which is the part of the premise the game is actually about — and it clears the search collision the verb form carries.
- **The game is single-player.** The first draft opened "1–4 player co-op" and treated that as the premise rather than a mode. It is now **single-player first, with co-op as a later multiplier on a proven loop** — and this is the largest scope reduction the project has taken, larger than cutting three classes. It does three things at once. It *retires* the replication risk the first draft called "the second-biggest risk," rather than deferring it with a cost. It cuts the entity gate by roughly 4× (\$5.1). And it *answers* a question the premise had left open — whether uniting flames was a co-op mechanic or a run objective. It is

the run objective: the other bearers are found in the dark over the course of a run. That is the better version, because the objective is now always present instead of contingent on having three friends online.

- **The roster narrows to one class.** The first draft scoped four classes for v1 and one for the slice; this draft commits the semester to the **Vanguard** alone and says so everywhere (§3.6, §5).
- **Ad-hoc dispatch was replaced with a scored, lock-checked backlog.** The reason is a coordination failure specific to this project rather than a general tidiness urge — see §4.2.

3 Game Mechanics — What the Player Sees and Does

3.1 The moment

Top-down. You move one hero with WASD and a visible army moves with you. There is no unit selection, no drag-box, and no targeting: the entire command vocabulary is four keys, issuable instantly mid-fight, that state an *intent* the units interpret for themselves. That is not a simplification of an RTS — it is the fiction and the input model being the same decision. A god is obeyed in intent, not in orders.

KEY	STANCE	WHAT THE ARMY DOES
1	Follow	Formation rings around you; engage whatever comes close. The default.
2	Charge	Surge at the cursor. Wide engage range, +25% speed — thrown past the edge of your light, on faith.
3	Hold	Anchor the formation where it stands; short engage range. Asking people to stand at the edge of the light, where the dark presses.
4	Rally	Collapse tight onto you — 45% slot radius, short engage. Everyone inside the fire.

Built and playable. The same four are drivable headlessly via console for scripted measurement runs.

3.2 The leash — the rule everything else is built on

Every unit has a maximum distance from the hero, roughly 1.25 screens. Cross it and the unit latches "broken": it ignores your stance and behaves as Follow until it is well back inside — a hysteresis band, so a unit sitting exactly on the boundary cannot flicker between states. A warning fires at 80% of the radius, because breaking must never feel random, and the HUD reports how many units are currently outside.

Under the flame premise this stops being an invisible rule and becomes the most legible thing on screen: inside is lit ground and full-value units, outside is the dark ground state and units dimmed one palette step. The player reads their own reach at a glance.

The consequence is the design's load-bearing restriction: **you cannot park the army on a chokepoint and walk away.** Anchoring a line always means standing in the fight beside it. Hero relevance is enforced by rule rather than by tuning numbers until it happens to be true.

3.3 What a fight feels like

Damage arrives in visible blows, not as a slide. Each unit runs its own swing clock — windup, strike a third of the way in, recover — edge-triggered so a blow lands exactly once regardless of frame rate. The clocks are desynchronised at spawn; without that, a whole wave strikes on the same frame and the army reads as one pulsing organism rather than a crowd.

A struck unit flashes and is shoved 35uu — sized against measured spacing (86uu at rest, ~45uu compressed, 95uu melee reach) so it is visible without blowing the line apart. The flash is deliberately exempt from light attenuation: dimmed correctly it would quantise below the palette's lowest threshold and vanish at the edge of the fire, which is exactly where fighting starts.

3.4 The run, and how it is calibrated

Three seconds to deploy, then wave one, a breather in which your retinue refills toward its cap, wave two, another breather, wave three. Clear the last wave and you win the floor. Die and the run ends.

ZERO-INPUT BASELINE (HERO NEVER MOVES, NO STANCE EVER ISSUED)

	WAVE 1 SURVIVORS	WAVE 2 SURVIVORS	RESULT
Original continuous-damage model, 3 runs	97–103 / 120	4–12	Lost wave 3, by 4–13 enemies
Shipped defaults, 4 runs	109–111 / 120	16–24	Lost wave 3, by 131–145 enemies



Figure 2 — the same run, mid-fight. Retinue 78 of 120, hero at 248/500, brood 49 still alive. The six squad readouts have gone ragged — Shield down to 5/20, Vets 11/20 — which is what attrition looks like before the line breaks. Losing units is expected and mournable; it is why they have no fixed names.

The run is lost by default, in every run, with tight variance. That is the intended calibration and it is the reason this build is usable as an A/B harness: everything a player adds — repositioning, Rally before a flank lands, Hold on a choke, the hero's own damage aimed somewhere useful — is the margin between losing and winning. The remaining honest gap is that the current margin is 131–145 enemies rather than the original 4–13, so a player has more to claw back than the calibration intends. Closing it is another tuning pass, and it should be judged by feel as much as by the number, because the model underneath it changed (§2.2b).

3.5 Growth and upkeep — the progression axis

Retinue growth is the reward for a run; there is no loot system and its absence is a design position rather than an omission. Units are gained by rescuing and rallying what the dark has taken.

Size is governed, not capped. Every active unit draws upkeep from a per-run supply pool replenished only at supply sites. When demand outruns supply, units do not die — they **degrade**: dimmer, weaker, visibly unfed, and newly recruited units arrive degraded until supply recovers. Fire must be fed or it dims. One mechanic does three jobs: it gives the exponential power curve a real governor, it bounds the entity budget by keeping degraded units in the cheap simulation tier, and it gives pressure on the player somewhere to land that is not simply "more enemies." *Specced; not yet built.*

3.6 The Vanguard — the one class this semester ships

The many. You are the shield at the front and the banner they follow. Every cell door you break open, every conscript you free, your line grows longer, and by late run you are marching a liberated army through the dark. High count, disciplined; growth verb *rescue and rally*; a retinue of liberated soldiers and militia that reads as a marching legion.

Three further classes are designed and parked — a fortifier who awakens stone guardians, a skirmisher with a small elite pack, a healer trailing a constellation of kindled souls — chosen to occupy distinct corners of count-versus-quality so the massive-entity fantasy shows four faces. None of them are in scope for the remaining weeks (§5).



Figure 3 — the actual shipping atlas (`T_Swarm_2bit` , 8 rotations × 4 rows, imported and running). Brood read as a dark hooded mass with no discernible kit; retinue read pale, armoured, and *equipped* — shield, spear, banner. That separation is doing the work described in §3.7: at two hundred units you are not reading faces, you are reading light shape against dark shape.

3.7 How it looks, and why that is a systems decision

A locked four-value palette, flat and unlit, with a post-process dither. This is not only a production saving: at several hundred units, silhouette and value are the only channels that survive, so the palette is what keeps friend, foe and boss separable at a glance without health bars. The render budget goes to entity count rather than to shading. Exceptions to the four values exist, are deliberate, and are documented individually — the flame's white core and the hit-flash above are the two current ones, each with a stated reason.

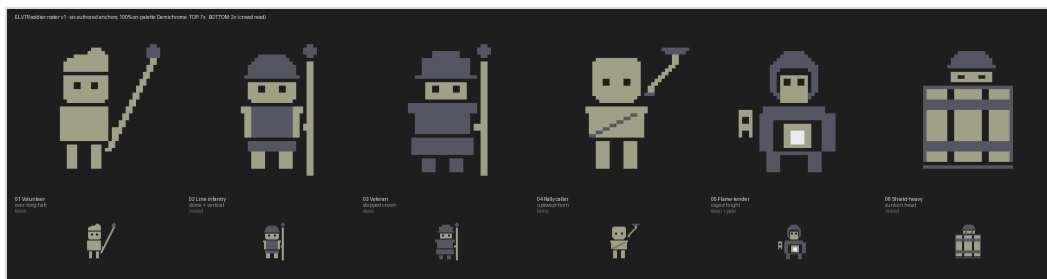


Figure 4 — the readability test the art has to pass. Six authored soldier anchors, 100% on the locked four-value ramp, shown at hero scale (top) and again at the size they actually occupy in a crowd (bottom). The top row is where you design; *the bottom row is where you judge*. Volunteer, line infantry, veteran, rally-caller, flame-tender and shield-heavy have to stay distinguishable at that second size or the silhouette language has failed — which is why every art spec is reviewed at gameplay zoom with hundreds of units moving, never as a single sprite on a canvas.

4 The Development Crew

The game is built by one person directing a crew of specialised agents, two MCP tool servers that give those agents real hands on the engine and the art pipeline, and a task system that keeps them from overwriting each other. Every canon file has **exactly one** agent with write access; everyone else treats it as read-only.

4.1 Roles — what each agent produces, and in what format

AGENT	WHAT IT PRODUCES	OUTPUT FORMAT & WHERE IT LANDS
narrative-director	The premise, decision-event text, and any fiction that survives \$2.1's test. Ends visual work with an art brief rather than describing an image.	Markdown prose in docs/narrative/ ; numbered briefs in docs/briefs/
pixel-art-director	Silhouette and value rules, palette tables, sprite-sheet layouts, readability reviews (Figure 4). Specs only — never image files.	Markdown specs in docs/art/ , each ending in an acceptance checklist
gameplay-director	Stat blocks per tier, the scaling curve, encounter budgets, retinue tuning — the numbers behind "a handful of militia" becoming an army.	Schema-validated JSON in docs/data/ that must import cleanly as an Unreal DataTable, plus a companion spec in docs/design/
performance-director	Frame-time measurement and the plan behind it: what we hold today versus what the design needs, never what we hope to hold.	Measured tables in docs/perf/BUDGETS.md , each row citing the harness that produced it
ui-director	HUD and menu layout, screen flow, widget specs — plus a viewable version before anything is built in engine.	Specs in docs/ui/ plus a self-contained HTML mockup published to a URL
host (backlog)	Sweeps the repo for undone and newly stale work; turns a goal into one scored, lock-checked task. Proposes and ranks; never approves, never spawns.	One task file per unit of work in docs/backlog/ , each carrying a paste-ready spawn prompt
Lead session	Briefs each director with only the section it owns, integrates the results, carries every dispatch and hand-back. The only role that spawns anything.	Commits, and the merged state of the canon
GDD Review Board	Six isolated-context reviewers — systems, narrative, player psychology, feasibility, adversarial QA, business — critique the canon independently, cross-examine, then a moderator synthesises.	A ranked, ratifiable edit plan; the party-vote rule in the first draft exists because this board argued for it
unreal-mcp (tool server)	Exposes the live Unreal 5.8 editor — actors, Blueprints, Niagara, materials — over HTTP, so an agent builds in-editor instead of describing a build.	Direct edits to the project; this is how the stances, the leash and the swarm exist at all
pixellab (tool server)	Generates sprites, portraits and animations to the art director's written spec.	PNGs to a holding folder, promoted only after a human keep/reject call

4.2 The two coordination rules that actually bite — and how they were mechanised

Both rules come from failures on this project, not from theory. Agents run as parallel sessions with their own context windows, which creates exactly two ways to lose work:

- **Two agents editing one file overwrite each other; there is no merge.** Every task declares the file globs it *owns*, and validation fails if two active tasks claim overlapping paths. The concrete case: five design specs each wanted to write the same systems document, so the shared write was split into its own task depending on all five. Specs fan wide; the file they converge on is a single join.
- **There is one Unreal editor.** Tasks declare the resources they need — editor, MCP port, art-generation credits — and those are global mutexes, so two agents can never be approved into driving the editor at once. Credits are locked because they cost real money.

Dispatch is gated on both checks and refuses any task not approved with its dependencies closed. The refusal is the point: **if it refuses, no tokens have been spent**. In practice this caps the crew at three or four live threads — a cap set by the single editor, not by ambition.

TWO FAILURE MODES WORTH RECORDING, BECAUSE BOTH COST TIME

An agent treats its spawn prompt as binding — telling a teammate mid-run that a restriction is lifted does not reliably override an instruction in its original prompt, and arguably shouldn't. The fix is to spawn a fresh one with the right scope, not to negotiate. Separately, an agent may honestly report that it has no access to a tool and then successfully call that same tool minutes later, because tools can be surfaced lazily rather than declared up front. Do not design around an agent's first answer about its own capabilities.

5 Scope and Constraints

5.1 Named constraints

CONSTRAINT	WHAT IT MEANS IN PRACTICE
One developer, one editor	All in-engine work serialises through a single live Unreal process. Directors write specs in parallel; exactly one thing builds at a time. Previously held in my head, now enforced by the resource mutex in §4.2.
The render bridge is over budget (the binding one)	14.62 ms of draw at 1,000 units against a 16.6 ms frame. The single-player decision <i>redefined</i> this gate — it is now one client at the late-run entity budget, with no replication term, roughly a quarter of the old bar — and the gate still does not pass . That is the clearest statement of where the project actually stands. The Niagara sprite path was repaired on 2026-07-26 but has no measured baseline yet, so the honest statement is that the replacement is untested rather than proven.
Art generation is separately metered	Sprite generation draws on a credit balance independent of language-model usage, so art volume is a hard-costed budget line. Every generation is retained before any keep/reject decision, so a rejected asset can still be compared against.

5.2 What one person can actually finish in the remaining weeks

The first draft projected a semester of one class, one biome, three floors, one boss, two decision events and two-to-four-player replication. **Four of those six are now out**, and one of them — the replication — is out permanently rather than postponed, because the game is single-player (§2.3). That correction is the scope lesson of this revision, and the sequencing below is ordered by what invalidates what.

#	IN SCOPE	WHY IT IS FIRST
1	Measure the repaired sprite path; record a GO / ADJUST / KILL verdict on the entity gate	Nothing downstream is trustworthy until this is a number rather than a hope. A KILL invalidates the concept rather than trimming it; the fallbacks — fewer entities, or a faked crowd — are already named.
2	Play the prototype and answer its five open feel questions	Is Hold a real tool at this leash radius? Does a leash break read, or does the army feel disobedient? Is Charge distinct from Follow? Does the hero feel like a commander at reduced damage? Does 120 units read at this camera height? None are settled by any number above.
3	Render the leash as light	The cheapest thing the reset unlocked, and the direct fix for question 2 — a floor decal and a value shift on data the renderer already has.
4	Sprites in place of debug boxes (Figure 3 is the atlas; it is imported and running)	Gated on 1. The current renderer is both over budget and visually placeholder; one change addresses both.
5	One hand-assembled arena floor with a boss, replacing ring-spawn	Turns a wave harness into a run. Hand-built deliberately — procedural generation is out.

EXPLICITLY OUT, AND WHY

The other three classes — one class is enough to prove the command layer, and four is four times the art. **Procedural generation** — the fallback to hand-assembled floors is already written into the slice definition, and constraint 5.1 says I get to build one thing at a time. **Persistent world state** — discarded with the lore in §2.1 and correctly not rewritten yet. **Loot** — deferred by design, because retinue growth is the reward loop. **Networked co-op** — not a semester cut but a *design decision* (§2.3): the game is single-player, and four-client replication of aggregate swarm state is not a thing this project has to prove at all. The design keeps the door open at zero cost rather than walking through it — a stance is one intent enum per army rather than per unit, and the leash clusters units on their hero, which is exactly what would make replication relevancy tractable if co-op ever returns.

6 Appendix — Build Status

What separates this document from the first draft is that most of the following can be checked by running the project rather than by trusting the prose — Figures 1 and 2 are that build.

SYSTEM	STATE	EVIDENCE
Four stances + hero movement	Built	Playable on keys 1–4; also drivable from console for scripted runs
Leash — radius, hysteresis, warn bit, break-to-Follow	Built	Per-unit latch; live broken count on the HUD
Mass Entity swarm — grid, steering, separation	Built	Measured: 120 of 120 units at distinct positions, 86uu median spacing at rest, no collision system involved
Discrete-swing combat, hit flash, knockback	Built	Tuned across five measured model variants (§2.2b)
Three-wave run with win and lose states	Built	Zero-input baseline loses repeatably (§3.4)
Four-value palette + post-process dither	Built	Locked ramp; documented exceptions
Niagara sprite rendering	Repaired, unmeasured	Root cause was an emitter simulating on GPU where it needed CPU; no performance baseline yet
Leash rendered as light	Specced	§2.1 — next build item
Upkeep economy	Specced	§3.5
Decision events	Specced	Templates discarded with the lore; the resolution rule survives
Procedural generation, loot, persistence, networking	Out of scope	§5.2

Kindled — Final Game Design Document · Aaron Low · 27 July 2026 · Multi-Agent AI for Game Development, Assignment #02. Working title; supersedes the Assignment #1 draft submitted as *Emberkeep*. All measurements cited were produced by the project’s own in-engine benchmark harness on the current development branch, single client.